

UNITED STATES PATENT APPLICATION

APPLICANT / ASSIGNEE: B-Lawz Music LLC

COUNTRY OF RESIDENCE: United States of America

ATTORNEY DOCKET NO.: BLM-001

TITLE OF INVENTION

TECHNICAL FIELD

The present invention relates to machine learning model training, and more particularly to systems and methods for compressing the effective training time of diffusion-based generative models running on central processing unit (CPU) hardware through the simultaneous application of five orthogonal acceleration techniques combined with a simulated experience clock that tracks year-equivalent training throughput.

BACKGROUND OF THE INVENTION

Training large-scale diffusion models conventionally requires graphics processing unit (GPU) clusters operating for days or weeks. Organizations lacking access to such infrastructure – including independent creators, small studios, and domain-specific AI deployments – are effectively locked out of training or fine-tuning diffusion models on proprietary datasets. The throughput gap between a modern A100 GPU cluster (approximately 180 gradient steps per second per GPU) and a commodity CPU (approximately 4.5 steps per second) represents a compression ratio of roughly 40:1 before any acceleration is applied.

Prior art approaches to this problem address individual acceleration vectors in isolation: data augmentation improves sample efficiency but does not address optimizer stalling; curriculum learning improves convergence trajectories but does not increase per-step informational density; experience replay from prioritized memory buffers reduces catastrophic forgetting but is not combined with online interpolation or adaptive learning rate surgery. No prior system simultaneously applies all five of these orthogonal techniques in a unified training loop, nor does any prior system define a hardware-agnostic "simulated experience clock" that quantifies multi-year

training throughput in terms of actual elapsed wall-clock minutes.

There exists a need for a unified training acceleration engine that (1) applies multiple orthogonal compression techniques simultaneously, (2) provides a mathematically grounded equivalence metric between CPU training time and GPU training time, and (3) exposes a real-time simulated experience clock that allows operators to reason about domain expertise accumulation independent of the underlying hardware.

SUMMARY OF THE INVENTION

The present invention provides a training time compression system comprising five simultaneously active acceleration modules orchestrated by a unified session controller, combined with a year-equivalent (YE) throughput engine that tracks effective training progress in units of simulated years of experience rather than raw gradient steps.

In a first aspect, the invention provides a method of training a diffusion model on CPU hardware comprising: (a) receiving a training frame from a domain-specific dataset; (b) generating a burst of N augmented variants of said frame through stochastic spatial transformations, each variant contributing its gradient to a shared weight update; (c) synthesizing interpolated frames between pairs of real frames by linearly blending frame tensors and mixing their conditioning prompts; (d) monitoring a rolling loss history and applying a surgical learning rate boost when loss slope falls below a plateau threshold for a configurable number of steps; (e) sorting training examples by a difficulty metric derived from per-scene loss variance and feeding them to the model in an easy-to-hard curriculum order; (f) pairing consecutive training frames as $(t, t+1)$ inputs with a temporal coherence penalty; and (g) computing a year-equivalent step count by weighting each step type by its informational density

relative to conventional single-frame CPU training.

In a second aspect, the invention provides a simulated experience clock operating at a fixed conversion rate of one real wall-clock minute equals one simulated year of training experience, where said conversion rate reflects the compound acceleration of all five techniques applied simultaneously.

In a third aspect, the invention provides a year-equivalent deficit engine that computes the gap between actual YE throughput and the target pace of one simulated year per real minute, and uses this deficit to drive downstream systems including replay cycle scheduling and training pressure computation for upstream quality gate controllers.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

I. System Architecture Overview

The training time compression system (hereinafter "the System") comprises the following functional modules operating in a coordinated training loop:

- Module A – Augmentation Burst Engine
- Module B – Scene Interpolation Synthesizer
- Module C – Adaptive Learning Rate Surgeon
- Module D – Curriculum Phase Controller
- Module E – Temporal Consistency Pair Generator
- Module F – Year-Equivalent Throughput Engine (the "Experience Clock")
- Module G – Session Registry and Reporting Layer

II. Module A – Augmentation Burst Engine

The Augmentation Burst Engine receives a single real training frame f and produces a burst of B variants $\{f_1, f_2, \dots, f_B\}$ through stochastic spatial transformations. Each variant

undergoes an independent random subset of: horizontal flip, vertical flip, random crop with resize, brightness jitter in $[0.85, 1.15]$, contrast jitter in $[0.85, 1.15]$, hue rotation in $[-0.05, +0.05]$, and Gaussian noise injection at $\sigma \in [0.01, 0.03]$.

The gradients of all B variants are accumulated before a single parameter update is applied. This produces an effective batch diversity equivalent to training on B distinct frames while paying the I/O cost of loading only one frame. Each burst variant earns `_BURST_YEAR_WEIGHT` = 6.0 year-equivalent steps, reflecting that six diverse gradient directions are collapsed into a single efficient update.

In the preferred embodiment, `burst_size` B = 8, yielding a per-frame informational density of $8 \times 6 = 48$ YE-steps per real training frame loaded.

III. Module B – Scene Interpolation Synthesizer

The Scene Interpolation Synthesizer generates synthetic training examples by linearly interpolating between two real frames `f_a` and `f_b` sampled from the dataset. The blended frame `f_blend` = $\alpha \times f_a + (1 - \alpha) \times f_b$ for $\alpha \in [0, 1]$ sampled uniformly. The conditioning prompt for the blended frame is constructed by selecting the first sentence of `prompt_a` followed by the final clause of `prompt_b`, weighted by α .

This technique expands the effective dataset size without requiring additional real data retrieval. Interpolated frames are injected into the training batch at a configurable density `interp_density` $\in [0.0, 0.8]$ of steps per epoch. Each interpolated frame earns `_INTERP_YEAR_WEIGHT` = 3.0 year-equivalent steps.

IV. Module C – Adaptive Learning Rate Surgeon

The Adaptive LR Surgeon monitors a rolling window of loss values of configurable length `lr_adapt_window` (default 40 steps) and

computes the linear slope of the loss trend using least-squares regression. When the slope exceeds a plateau threshold (i.e., loss is flat or rising) for `plateau_patience` consecutive steps, the surgeon applies a multiplicative boost of `lr_boost_factor` (default 1.8) to the current learning rate.

When loss subsequently resumes improvement, the boost is removed and the base schedule is restored. If the boosted LR causes loss to worsen, a decay factor `lr_decay_factor` (default 0.85) is applied. This mechanism prevents the optimizer from stalling in flat loss basins without requiring manual hyperparameter tuning.

The surgeon operates without modifying the base learning rate schedule – it applies a transient multiplicative correction that is removed as soon as gradient flow is restored. This makes it safe to combine with any existing LR scheduler.

V. Module D – Curriculum Phase Controller

The Curriculum Phase Controller maintains a per-scene loss map `{scene → [loss1, loss2, ..., lossN]}` updated after each training step. The difficulty of a scene is defined as the variance of its recent loss values: high variance indicates a scene the model has not yet converged on. At the start of each epoch the full dataset is sorted by scene difficulty in ascending order, presenting the model with easy (low-variance, near-converged) scenes first and hard (high-variance, still learning) scenes last.

This curriculum ordering mirrors the progressive training strategy used in large-scale GPU training pipelines and has been shown to accelerate convergence by preventing the optimizer from oscillating between easy and hard examples.

VI. Module E – Temporal Consistency Pair Generator

The Temporal Consistency Pair Generator selects pairs of consecutive frames (`ft`, `ft+1`) from the same video sequence

and trains the model on both simultaneously with a coherence penalty term added to the loss:

$$L_{\text{total}} = L_{\text{reconstruction}}(f_t) + L_{\text{reconstruction}}(f_{t+1}) + \lambda \times ||\text{latent}(f_t) - \text{latent}(f_{t+1})||^2$$

where λ is a configurable coherence weight and $\text{latent}(f)$ denotes the model's internal latent representation of frame f . This teaches the model that adjacent frames must be perceptually related, building temporal consistency before full temporal UNet training is applied.

VII. Module F – Year-Equivalent Throughput Engine

The Year-Equivalent Throughput Engine (the "Experience Clock") is the central novelty of the present invention. It defines a hardware-agnostic metric of training progress called the year-equivalent (YE) step count.

The target pace is derived from the baseline throughput of an 8-core CPU:

$$\begin{aligned} _YEAR_EQUIV_STEPS_PER_MINUTE &= _CPU_STEPS_PER_SEC_BASELINE \times \\ &365.25 \times 24 \times 3600 \\ &= 4.5 \times 31,557,600 \\ &\approx 142,009,200 \text{ YE-steps per real minute} \end{aligned}$$

This figure represents how many gradient steps a conventional single-frame CPU training loop would execute in one real minute over one calendar year of continuous operation. The Experience Clock therefore defines "one simulated year" as the equivalent informational throughput of one calendar year of naive CPU training.

The Experience Clock maintains a running total `_year_equiv_steps` and three counters for each step type, weighted as follows:

$$\text{Burst variant step: } _BURST_YEAR_WEIGHT = 6.0 \text{ YE-steps}$$

Priority replay step: `_REPLAY_YEAR_WEIGHT = 12.0` YE-steps

Synthetic interpolation frame: `_INTERP_YEAR_WEIGHT = 3.0`
YE-steps

Scenario-sourced step (depth 0): `_SCENARIO_YEAR_WEIGHT_BASE = 18` YE-steps

Scenario-sourced step (depth 1):
`_SCENARIO_YEAR_WEIGHT_COMPOUND_1 = 24` YE-steps

Scenario-sourced step (depth
2+): `_SCENARIO_YEAR_WEIGHT_COMPOUND_2 = 30` YE-steps

The year-equivalent deficit at elapsed time t is:

$$\text{deficit}(t) = \max(0, \text{target_pace} \times t_{\text{minutes}} - \text{_year_equiv_steps})$$

This deficit is consumed by two downstream systems: (1) the replay cycle scheduler, which computes `recommended_replay_cycles = deficit / (REPLAY_BATCH_SIZE × _REPLAY_YEAR_WEIGHT)`, and (2) the Caffeine Mode pressure signal consumed by the content quality gate and scenario selection layer, which converts the deficit into a dimensionless pressure score on $[0, 1.5]$.

VIII. Module 6 – Session Registry

The Session Registry records per-session and cumulative training statistics including: `simulated_years`, `simulated_experience` (human-readable), `real_elapsed`, `effective_steps`, `lr_boosts`, `lr_decays`, `interp_generated`, `burst_calls`, and `curriculum_phases`. These statistics are serialized to JSON and exposed via a FastAPI `/train/simulator/status` endpoint.

IX. 10-Year / 10-Minute Session Target

In the preferred embodiment, the System is calibrated so that each training session completes in approximately 10 real minutes and accumulates approximately 10 simulated years of training

experience. The MaxCore dataset bridge refreshes its prompt pool every 10 minutes, synchronized with each session, so each 10-year session trains on a freshly sampled, randomly ordered set of domain-specific prompts. After each session, updated model weights are distributed to all connected client nodes.

X. Interoperability with Scenario Engine and Memory System

The YE deficit output of Module F is consumed by:

- The Music Industry Compounding Scenario Engine (see Patent No. 2), which uses compound_depth-weighted YE credits
- The A/B Test Scenario Layer (see Patent No. 3), which converts the deficit into training pressure for UCB1 bandit exploration-exploitation tuning
- The Four-Tier Memory Replay System (see Patent No. 5), which uses the recommended_replay_cycles count to drive post-epoch memory playback

CLAIMS

1. A computer-implemented system for training a generative diffusion model on central processing unit hardware, the system comprising: a processor; and a non-transitory computer-readable medium storing instructions that, when executed by the processor, cause the processor to: receive a training frame from a dataset; generate a plurality of augmented variants of the training frame using stochastic spatial transformations and accumulate gradients of all variants before applying a single weight update; synthesize interpolated training frames by blending pairs of real frames at random interpolation weights and constructing blended conditioning prompts; monitor a rolling loss history and apply a multiplicative learning rate boost when the loss slope falls below a plateau threshold for a configurable patience count; sort training examples by per-scene loss variance to produce a curriculum ordering from easy to hard within each training epoch; pair consecutive training frames with a temporal coherence penalty applied to the summed reconstruction loss; and maintain a year-equivalent step counter that assigns a distinct informational density weight to each step type and computes an accumulated simulated training experience in units of years.
2. The system of claim 1, wherein the year-equivalent step counter defines a conversion rate of one real wall-clock minute equal to one simulated year of training experience, derived from the baseline throughput of a commodity central processing unit operating for one calendar year of continuous single-frame training.
3. The system of claim 1, wherein the year-equivalent step counter assigns a first weight to augmentation burst variant steps, a second weight greater than the first weight to priority experience replay steps, a third

weight less than the first weight to synthetic interpolated frame steps, and a fourth weight greater than the second weight to scenario-sourced domain-specific steps.

4. The system of claim 3, wherein the fourth weight varies by scenario compound depth, assigning an escalating weight to scenario steps at depth zero, depth one, and depth two or greater, respectively, reflecting increasing informational density of compounded narrative training signals.
5. The system of claim 1, further comprising a year-equivalent deficit engine that computes, at each elapsed real time interval, the difference between a target year-equivalent step count and the accumulated year-equivalent step count, and exposes the deficit as a training pressure signal to at least one downstream system.
6. The system of claim 5, wherein the at least one downstream system is a quality gate controller that uses the training pressure signal to relax a content intensity threshold when the deficit exceeds a configurable critical threshold.
7. The system of claim 5, wherein the at least one downstream system is a bandit-based scenario selection layer that uses the training pressure signal to increase exploitation of high-yield training domains over exploration of unknown training domains.
8. The system of claim 5, wherein the at least one downstream system is a memory replay scheduler that uses the deficit to compute a recommended number of post-epoch replay cycles to close the informational density gap.

9. The system of claim 1, wherein the multiplicative learning rate boost is a transient correction applied without modifying the base learning rate schedule, and wherein the boost is removed when the loss resumes improvement below the plateau threshold.
10. The system of claim 1, wherein the per-scene loss variance is maintained in a per-scene loss map updated after each training step, and wherein the curriculum ordering is recomputed at the start of each training epoch.
11. The system of claim 1, wherein the temporal coherence penalty is computed as the L2 norm of the difference between the internal latent representations of two consecutive frames.
12. The system of claim 1, further comprising a session registry that records per-session and cumulative training statistics including simulated years of experience, and exposes said statistics via a network-accessible status endpoint.
13. A method of measuring effective training progress of a machine learning model training session, the method comprising: defining a year-equivalent step unit representing the informational density of one conventional single-frame training step on baseline hardware operating for one calendar year; assigning distinct year-equivalent weights to each of a plurality of step types including data augmentation burst steps, experience replay steps, and synthetic interpolation steps; accumulating a running total of year-equivalent steps weighted by step type; computing a target year-equivalent step count based on elapsed real time at a fixed conversion rate; and reporting a deficit representing the difference between the target and

accumulated year-equivalent step counts.

14. The method of claim 13, wherein the fixed conversion rate is one simulated year of training experience per one real wall-clock minute.
15. The method of claim 13, further comprising converting the deficit into a dimensionless pressure score on a range from zero to a maximum pressure value and providing said pressure score to at least one downstream training control system.
16. The method of claim 13, wherein assigning distinct year-equivalent weights comprises assigning a higher weight to steps whose gradients carry domain-specific structured narrative context than to steps whose gradients are derived from stochastic augmentation alone.
17. A computer-implemented method for accelerating diffusion model fine-tuning on a central processing unit, the method comprising: simultaneously applying, within a single training loop, five orthogonal acceleration techniques comprising: augmentation burst training, synthetic scene interpolation, adaptive learning rate surgery, curriculum-ordered scene presentation, and temporal consistency pair training; maintaining a unified year-equivalent throughput counter that aggregates contributions from all five techniques using distinct informational density weights; and computing a simulated training experience metric in units of calendar years from the unified counter.
18. The method of claim 17, further comprising distributing updated model weights to client nodes after each training session at a synchronization interval matching a domain-specific dataset refresh cycle.

- 19.** The method of claim 17, wherein the simultaneous application of the five techniques achieves an effective compression ratio of approximately ten real hours of GPU training per one real hour of central processing unit training.
- 20.** A non-transitory computer-readable medium storing instructions that, when executed by a processor, implement: an augmentation burst engine configured to generate a configurable number of stochastically augmented variants of each training frame and accumulate their gradients before a weight update; an interpolation synthesizer configured to blend pairs of real training frames and their conditioning prompts at random interpolation weights to produce synthetic training samples; an adaptive learning rate surgeon configured to monitor a rolling loss window and apply a transient multiplicative learning rate correction when the loss slope indicates a plateau; a curriculum phase controller configured to sort training examples by per-scene loss variance in ascending order at the beginning of each training epoch; a temporal consistency pair generator configured to impose a coherence penalty on consecutive training frame pairs; and a year-equivalent experience clock configured to assign distinct informational density weights to each step type and report accumulated simulated training experience in units of years.

ABSTRACT

A training time compression system for diffusion model fine-tuning on CPU hardware simultaneously applies five orthogonal acceleration techniques: augmentation burst training (multiple stochastic variants per frame), synthetic scene interpolation (blended frame-prompt pairs), adaptive learning rate surgery (transient plateau-breaking boosts), curriculum-ordered scene presentation (easy-to-hard difficulty sorting), and temporal consistency pair training (consecutive-frame coherence penalties). A year-equivalent (YE) throughput engine assigns distinct informational density weights to each step type and maintains a running total in units of simulated years of training experience, operating at a fixed conversion rate of one simulated year per one real wall-clock minute. A deficit engine computes the gap between target and actual YE throughput and exposes a dimensionless training pressure signal to downstream quality gate and scenario selection systems. Updated model weights are distributed to all connected nodes after each 10-minute session, which accumulates approximately 10 simulated years of domain-specific training experience.